

AQUÍ VA EL TÍTULO DEL ARTÍCULO

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ABSTRACT. Aquí va el resumen

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1. INTRODUCTION

[PP21, SS06a, Hoc69]

2. PRELIMINARIES

[Sim06a, Sex03, Joh82]

3. COMPACT QUOTIENTS

Necesitamos quitar toda esta sección.

Aquí hay
que poner la
referencia.

Let $f: A \rightarrow B$ be a frame morphism with right adjoint $f_*: B \rightarrow A$. If $F \subseteq A$ and $G \subseteq B$ are filters, we define:

$$(1) \quad b \in f^*F \iff f_*(b) \in F \quad a \in f_*G \iff f(a) \in G,$$

where f^*F and f_*G are filters in B and A , respectively.

Proposition 3.1. For $f = f^*: A \rightarrow B$ a frame morphism and $G \in B^\wedge$, then $f_*G \in A^\wedge$.

Proof. By (1), f_*G is a filter in A . We need that f_*G satisfy the open filter condition. Let $X \subseteq A$ be such that $\bigvee X \in f_*G$, with X directed. Then

$$Y = \{f(x) \mid x \in X\}$$

is directed and $f(\bigvee X) = \bigvee f[X] = \bigvee Y \in G$. Since G is an open filter and $f(\bigvee X) = \bigvee f[X] \in G$, there exists $x \in X$ such that $f(x) \in G$. Therefore, $x \in f_*G$. $\square$

Simmons shows in [Sim04, Lemma 8.9 and Corollary 8.10] that the following diagram

$$(2) \quad \begin{array}{ccc} A & \xrightarrow{f^\infty} & A \\ U_A \downarrow & & \downarrow U_A \\ \mathcal{O}S & \xrightarrow{F^\infty} & \mathcal{O}S \end{array}$$

commutes laxly, that is, $U_A \circ f^\infty \leq F^\infty \circ U_A$. In this diagram f^∞ and F^∞ represent the fitted nuclei associated with the filters $F \in A^\wedge$ and $\nabla \in \mathcal{O}S^\wedge$, respectively, that is,

$$f^\infty = v_F \quad \text{and} \quad F^\infty = v_\nabla.$$

Here we prove something more general, since we consider $j \in NA$, $F \in A_j^\wedge$ and $j_*F \in A^\wedge$ to produce the diagram

$$\begin{array}{ccc} A & \xrightarrow{f^\infty} & A \\ j^* \downarrow & & \downarrow j^* \\ A_j & \xrightarrow{\hat{f}^\infty} & A_j \end{array}$$

where $\hat{f}^\infty$ and f^∞ are the fitted nuclei associated with the filters j_*F and F , respectively.

The following lemma is part of the analysis developed in this work and constitutes a key tool in the construction of the commutative diagrams that follow.

Lemma 3.2. *For $j = j^*$, f and $\hat{f}$ as before, it holds that $j \circ \hat{f} \leq f \circ j$.*

Proof. Recall that, by (1)

$$\hat{f} = \bigvee \{v_y \mid y \in j_*F\} \quad \text{y} \quad f = \bigvee \{v_{j(y)} \mid j(y) \in F\}.$$

Let $a \in A$ be. For every $y \in j_*F$, we have

$$j(v_y(a)) = j(y \succ a) \leq (j(y) \succ j(a)) = v_{j(y)}(j(a)) \leq f(j(a)),$$

where the last inequality follows from $j(y) \in F$. Therefore, since j preserves arbitrary joins,

$$j(\hat{f}(a)) = j\left(\bigvee \{v_y(a) \mid y \in j_*F\}\right) = \bigvee \{j(v_y(a)) \mid y \in j_*F\} \leq f(j(a)).$$

Hence, $j \circ \hat{f} \leq f \circ j$. $\square$

We now prove the above for an arbitrary ordinal.

Corollary 3.3. *For j , f and $\hat{f}$ as before, it holds that $j \circ \hat{f}^\alpha \leq f^\alpha \circ j$ where $\alpha \in \text{Ord}$.*

Proof. We proceed by transfinite induction on α . If $\alpha = 0$, the result is immediate.

For the induction step, suppose that the inequality holds for α . Then

$$j \circ \hat{f}^{\alpha+1} = j \circ \hat{f} \circ \hat{f}^\alpha \leq f \circ j \circ \hat{f}^\alpha \leq f \circ f^\alpha \circ j = f^{\alpha+1} \circ j,$$

where the first inequality follows from Lemma 3.2 and the second from the induction hypothesis.

Finally, let λ be a limit ordinal. For every $a \in A$,

$$j(\hat{f}(a)) = j\left(\bigvee_{\alpha < \lambda} \hat{f}^\alpha\right) = \bigvee_{\alpha < \lambda} j(\hat{f}^\alpha(a)) \leq \bigvee_{\alpha < \lambda} f^\alpha(j(a)) = f^\lambda(j(a)).$$

where the inequality follows from the induction hypothesis. Hence

$$j \circ \hat{f}^\lambda \leq f^\lambda \circ j.$$

This completes the proof. $\square$

Recalling that $v_F = f^\infty$ and $v_{j_*F} = \hat{f}^\infty$, Corollary 3.3 yields

$$j \circ v_{j_*F} \leq v_F \circ j$$

We now compare the compact quotients induced by F and j_*F , and construct a frame morphism between them making the corresponding diagram commute.

If we consider j , v_F and v_{j_*F} we have the following diagram

$$(3) \quad \begin{array}{ccc} A & \xrightleftharpoons[(v_{j_*F})_*]{(v_{j_*F})^*} & A_{j_*F} \\ \downarrow j & \searrow H & \\ A_j & \xrightleftharpoons[(v_F)^*]{(v_F)_*} & A_F \end{array}$$

where A_F and A_{j_*F} are the compact quotients produced by v_F and v_{j_*F} , respectively.

Define

$$H: A \rightarrow A_F, \quad H = v_F \circ j.$$

Since v_F is idempotent, $H(A) \subseteq A_F$. Hence, by restricting H to A_{j_*F} , we obtain a map

$$h: A_{j_*F} \rightarrow A_F, \quad h = H|_{A_{j_*F}}.$$

As finite meets in quotient frames are inherited from the ambient frame, h preserves finite meets. It remains to verify that h preserves arbitrary joins.

Lemma 3.4. *The map $h: A_{j_*F} \rightarrow A_F$ preserves arbitrary joins.*

Proof. Let $X \subseteq A_{j_*F}$. Since joins in A_{j_*F} are given by

$$\bigvee_{v_{j_*F}} X = v_{j_*F}(\bigvee X)$$

we have

$$\begin{aligned} h\left(\bigvee_{v_{j_*F}} X\right) &= v_F\left(j\left(v_{j_*F}\left(\bigvee X\right)\right)\right) \\ &\leq v_F\left(v_F\left(j\left(\bigvee X\right)\right)\right) \\ &= v_F\left(j\left(\bigvee X\right)\right), \end{aligned}$$

where the inequality follows from Corollary 3.3 and the equality from the idempotency of v_F .

Since H is a frame morphism,

$$v_F\left(j\left(\bigvee X\right)\right) = \bigvee_{v_F}\{H(x) \mid x \in X\} = \bigvee_{v_F}\{h(x) \mid x \in X\}.$$

Therefore,

$$h\left(\bigvee_{v_{j_*F}} X\right) \leq \bigvee_{v_F} h[X].$$

Conversely, for every $x \in X$,

$$x \leq \bigvee_{v_{j_*F}} X,$$

and hence, by monotonicity of h ,

$$h(x) \leq h\left(\bigvee_{v_{j_*F}} X\right).$$

Since this holds for every $x \in X$, we have

$$\bigvee_{v_F} h[X] \leq h\left(\bigvee_{v_{j_*F}} X\right).$$

Thus,

$$h\left(\bigvee_{v_{j_*F}} X\right) = \bigvee_{v_F} h[X].$$

Therefore h preserves arbitrary joins. $\square$

The previous construction yields the following commutative diagram induced by the compact quotients associated with the fitted nuclei.

Proposition 3.5. *The diagram*

$$(4) \quad \begin{array}{ccc} A & \xrightarrow{v_{j^*F}} & A_{j^*F} \\ j \downarrow & & \downarrow h \\ A_j & \xrightarrow{v_F} & A_F \end{array}$$

is commutative.

Proof. By Lemma 3.4, h is a frame morphism. The commutativity of the diagram follows from the definition of h . $\square$

A particular instance of diagram (4) is obtained by taking $j^* = \mathcal{U}_A$, where $S = \text{pt } A$. Thus, we obtain the following commutative diagram:

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ \mathcal{U}_A \downarrow & & \downarrow h \\ \mathcal{O}S & \xrightarrow{v_\nabla} & \mathcal{O}S_\nabla \end{array}$$

Recall that $\text{pt } A_F = Q$. Hence, the spatial reflection of A_F is given by

$$\mathcal{U}_{A_F}: A_F \rightarrow \mathcal{O} \text{pt } A_F = \mathcal{O}Q.$$

Similarly, since $\text{pt}((\mathcal{O}S)_\nabla) = Q$, we have the spatial reflection

$$\mathcal{U}_{\mathcal{O}S_\nabla}: \mathcal{O}S_\nabla \rightarrow \mathcal{O} \text{pt}(\mathcal{O}S)_\nabla = \mathcal{O}Q.$$

Consequently, the two induced morphisms from A to $\mathcal{O}Q$ are both given by

$$x \mapsto \mathcal{U}_A(x) \cap Q.$$

Therefore, the preceding constructions fit into the following commutative diagram:

$$(5) \quad \begin{array}{ccccc} A & \xrightarrow{v_F} & A_F & & \\ \mathcal{U}_A \downarrow & & \downarrow h & \searrow \mathcal{U}_{A_F} & \\ & & & & \mathcal{O}Q \\ \mathcal{O}S & \xrightarrow{v_\nabla} & \mathcal{O}S_\nabla & \nearrow \mathcal{U}_{\mathcal{O}S_\nabla} & \end{array}$$

Since the above diagram is determined by $F \in A^\wedge$ and the associated compact saturated subset $Q \in \mathcal{Q}S$, we refer to diagram (5) as the *Q-square*.

Theorem 3.6. *Under the hypotheses of the Q-square, if A satisfies property (H), then*

$$\mathcal{O}Q \cong \uparrow Q'.$$

Equivalently, the frame of open sets of the point space of A_F is isomorphic to a compact closed quotient of a Hausdorff space.

Proof. Suppose that A satisfies **(H)**. Since **(H)** is conservative, $S = \text{pt } A$ is a T_2 space. Hence, for every $Q \in \mathcal{Q}S$, the set Q is closed, and therefore $Q' \in \mathcal{O}S$

Moreover, since every T_2 space is strongly stacked, we have

$$v_{\nabla} = [Q'].$$

As $Q' \in \mathcal{O}S$, it follows that

$$[Q'] = u_{Q'}$$

Consequently,

$$(\mathcal{O}S)_{v_{\nabla}} = (\mathcal{O}S)_{u_{Q'}} \cong \uparrow Q'.$$

Now, $u_{Q'}$ is a complemented nucleus in $\mathcal{O}S$, by [PP12, Proposition VI.3.3], implies that $\mathcal{O}S_{v_{\nabla}}$ is spatial. Therefore,

$$(\mathcal{O}S)_{v_{\nabla}} \cong \mathcal{O} \text{pt}((\mathcal{O}S)_{v_{\nabla}}).$$

Since

$$\text{pt}((\mathcal{O}S)_{v_{\nabla}}) = Q,$$

we obtain

$$\mathcal{O}Q \cong (\mathcal{O}S)_{v_{\nabla}} \cong \uparrow Q'.$$

□

4. TIDY FRAMES

Proposition 4.1. *If A is a tidy frame, then A_j is a tidy frame.*

Proof. Let A be α -tidy and let $F \in A^{\wedge}$. By Proposition 3.1, $j_*F \in A^{\wedge}$. Let

$$\hat{f} = \bigvee_{y \in j_*F} v_y \quad \text{and} \quad f = \bigvee_{x \in F} v_x.$$

be the prenuclei associated with j_*F and F , respectively.

Since A is tidy, there exists an ordinal α such that, for every $y \in j_*F$,

$$y \vee \hat{d} = 1, \quad \text{with } \hat{d} = \hat{f}^{\alpha}(0).$$

Let

$$d = f^{\alpha}(j(0)),$$

by Corollary 3.3,

$$j(\hat{d}) = j(\hat{f}^{\alpha}(0)) \leq f^{\alpha}(j(0)) = d.$$

Since j is inflationary, $\hat{d} \leq j(\hat{d})$, and hence

$$\hat{d} \leq d.$$

Now, let $x \in F$. Thus, $x \in A_j$, we have $j(x) = x$, and therefore $x \in j_*F$. Then,

$$x \vee \hat{d} = 1.$$

As $\hat{d} \leq d$, it follows that

$$1 = x \vee \hat{d} \leq x \vee d \leq 1.$$

Hence $x \vee d = 1$. Therefore, A_j is α -tidy. $\square$

The following theorem establishes that tidiness is preserved under binary coproducts.

Proposition 4.2. *Let A_1 and A_2 be α -tidy frames. Then $A_1 \oplus A_2$ is α -tidy.*

Proof. By construction of the coproduct $A_1 \oplus A_2$, consider the canonical frame morphism

$$\iota_i: A_i \rightarrow A_1 \oplus A_2, \quad i = 1, 2$$

together with their right adjoints $(\iota_i)_*: A_1 \oplus A_2 \rightarrow A_i$.

Then, we have the following diagram

$$\begin{array}{ccccccc} & & & \iota_i & & & \\ & & \swarrow & & \searrow & & \\ A_i & \xrightarrow{\alpha_i} & A_1 \times A_2 & \xrightarrow{\downarrow(-)} & \mathcal{L}(A_1 \times A_2) & \xrightarrow{\sim} & A_1 \oplus A_2 \\ & \nwarrow & & & & & \nwarrow \\ & & & (\iota_i)_* & & & \end{array}$$

By Proposition 3.1,

$$(\iota_i)_* F \in A_i^\wedge.$$

Since A_i is α -tidy, for every $x_i \in (\iota_i)_* F$ there exists $d_i \in A_i$ such that

$$x_i \vee d_i = 1,$$

where

$$d_i = f_i^\alpha(0) \quad \text{and} \quad f_i = \bigvee_{y \in (\iota_i)_* F} v_y.$$

By the definition of $(\iota_i)_* F$,

$$x_i \in (\iota_i)_* F \iff \iota_i(x_i) \in F.$$

Thus,

$$x_1 \oplus 1, 1 \oplus x_2 \in F.$$

Since F is a filter,

$$(x_1 \oplus 1) \wedge (1 \oplus x_2) = x_1 \oplus x_2 \in F.$$

Now let

$$f_F = \bigvee_{z \in F} v_z$$

be the prenucleus associated with F . We first show that, for $i = 1, 2$,

$$\iota_i \circ f_i \leq f_F \circ \iota_i.$$

Indeed, let $a \in A_i$ and $y \in (\iota_i)_* F$. Since $\iota_i(y) \in F$,

$$\begin{aligned} \iota_i(v_y(a)) &= \iota_i(y \succ a) \\ &\leq (\iota_i(y) \succ \iota_i(a)) \\ &= v_{\iota_i(y)}(\iota_i(a)) \\ &\leq f_F(\iota_i(a)). \end{aligned}$$

Therefore, since the join defining f_i is pointwise and ι_i preserves arbitrary joins,

$$\iota_i(f_i(a)) = \iota_i\left(\bigvee_{y \in (\iota_i)_* F} v_y(a)\right) = \bigvee_{y \in (\iota_i)_* F} \iota_i(v_y(a)) \leq f_F(\iota_i(a)).$$

Hence,

$$\iota_i \circ f_i \leq f_F \circ \iota_i.$$

We now prove by transfinite induction that

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal β .

If $\beta = 0$, the result is immediate. Suppose that it holds for β . Then

$$\begin{aligned} \iota_i \circ f_i^{\beta+1} &= \iota_i \circ f_i \circ f_i^\beta \\ &\leq f_F \circ \iota_i \circ f_i^\beta \\ &\leq f_F \circ f_F^\beta \circ \iota_i \\ &= f_F^{\beta+1} \circ \iota_i. \end{aligned}$$

Now let λ be a limit ordinal. For every $a \in A_i$,

$$\begin{aligned} \iota_i(f_i^\lambda(a)) &= \iota_i\left(\bigvee_{\beta < \lambda} f_i^\beta(a)\right) \\ &= \bigvee_{\beta < \lambda} \iota_i(f_i^\beta(a)) \\ &\leq \bigvee_{\beta < \lambda} f_F^\beta(\iota_i(a)) \\ &= f_F^\lambda(\iota_i(a)). \end{aligned}$$

Thus,

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal β .

Evaluating at $0 \in A_i$ for $\beta = \alpha$, we obtain

$$\iota_1(d_1) \leq f_F^\alpha(0 \oplus 1) \quad \text{and} \quad \iota_2(d_2) \leq f_F^\alpha(1 \oplus 0).$$

Therefore,

$$\begin{aligned} d_1 \oplus d_2 &= (d_1 \oplus 1) \wedge (1 \oplus d_2) \\ &\leq f_F^\alpha(0 \oplus 1) \wedge f_F^\alpha(1 \oplus 0) \\ &= f_F^\alpha(0 \oplus 1 \wedge 1 \oplus 0). \end{aligned}$$

Set $d = f_F^\alpha(0 \oplus 0)$, then $d_1 \oplus d_2 \leq d$. Finally,

$$\begin{aligned} 1 \oplus 1 &= (x_1 \vee d_1) \oplus (x_2 \vee d_2) \\ &= (x_1 \oplus x_2) \vee (d_1 \oplus d_2) \\ &\leq (x_1 \oplus x_2) \vee d \\ &\leq 1 \oplus 1. \end{aligned}$$

Therefore,

$$(x_q \oplus x_2) \vee d = 1 \oplus 1$$

and $A_1 \oplus A_2$ is α -tidy. $\square$

Corollary 4.3. *Let A_1 and A_2 be α -tidy and β -tidy, respectively. Then $A_1 \oplus A_2$ is γ -tidy, where*

$$\gamma = \max\{\alpha, \beta\}.$$

Proof. Let $\gamma = \max\{\alpha, \beta\}$. Then A_1 and A_2 are both γ -tidy. Hence, by Proposition 4.2, $A_1 \oplus A_2$ is γ -tidy. $\square$

We extend the previous result to arbitrary coproducts.

Proposition 4.4. *Tidiness is preserved under arbitrary coproducts.*

Proof. Let $\{A_i\}_{i \in I}$ be a family of tidy frames, and for each $i \in I$, let α_i be a tidiness degree of A_i . Consider

$$A = \bigoplus_{i \in I} A_i$$

and let

$$\iota_i: A_i \rightarrow A$$

denote the canonical coproduct morphisms. Since ι_i is a frame morphism, it has a right adjoint $(\iota_i)_*$.

Let $F \in A^\wedge$. By Proposition 3.1,

$$(\iota_i)_* F \in A_i^\wedge$$

for every $i \in I$. Moreover,

$$x_i \in (\iota_i)_* F \iff \iota_i(x_i) \in F.$$

Set $\alpha = \sup\{\alpha_i \mid i \in I\}$. Then every A_i is α -tidy. Hence, for every $x_i \in (\iota_i)_* F$,

$$x_i \vee d_i = 1,$$

where

$$d_i = f_i^\alpha(0) \quad \text{and} \quad f_i = \bigvee_{y \in (\iota_i)_* F} v_y.$$

Let

$$f_F = \bigvee_x \cdot \in Fv_x$$

be the prenucleus associated with F . As in the proof of Proposition 4.2, for every $i \in I$ we have

$$\iota_i \circ f_i \leq f_F \circ \iota_i,$$

and, by transfinite induction,

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal β .

Evaluating at $0 \in A_i$ for $\beta = \alpha$, we obtain

$$\iota_i(d_i) = \iota_i(f_i^\alpha(0)) \leq f_F^\alpha(\iota_i(0)) = f_F^\alpha(\langle 0 \rangle).$$

Set

$$d = f_F^\alpha(\langle 0 \rangle).$$

Therefore,

$$\iota_i(d_i) \leq d$$

for every $i \in I$. Taking arbitrary joins yields

$$\langle d_i \rangle \bigvee_{i \in I} \iota_i(d_i) \leq d.$$

Now let

$$\langle x_i \rangle \in F.$$

Since $x_i \vee d_i = 1$ for every $i \in I$, we have

$$\langle x_i \rangle \vee \langle d_i \rangle = \langle x_i \vee d_i \rangle = \langle 1 \rangle.$$

As $\langle d_i \rangle \leq d$,

$$\langle 1 \rangle = \langle x_i \rangle \vee \langle d_i \rangle \leq \langle x_i \rangle \vee d \leq \langle 1 \rangle.$$

Thus,

$$\langle x_i \rangle \vee d = \langle 1 \rangle.$$

Therefore, $A = \bigoplus_{i \in I} A_i$ is α -tidy, and hence tidy. $\square$

Corollary 4.5. *The subcategory of tidy frames is a coreflective subcategory of $\mathbf{Frm}$.*

5. STACKED AND STRONGLY STACKED FRAMES

Lemma 5.1. *If $\alpha \in \mathbf{Ord}$ and $X \in CS$, then*

$$(6) \quad \partial_F^\alpha(X) = (f_F^\alpha(X'))'.$$

Proof. By transfinite induction on $\alpha \in \mathbf{Ord}$:

- If $\alpha = 0$, then $\partial_F^0 = \text{id}$, that is, $\partial_F^0(X) = X$ and $f_F^0 = \text{id}$. Thus

$$\partial_F^0(X) = X = (X'') = (f_F^0(X'))'.$$

- Assume that $\partial_F^\alpha(X) = (f_F^\alpha(X'))'$ holds. By definition,

$$\partial_F^{\alpha+1}(X) = \partial_F(\partial_F^\alpha(X)).$$

Furthermore,

$$\partial_F^{\alpha+1}(X) = (f_F((\partial_F^\alpha(X))'))'.$$

By induction hypothesis, $\partial_F^\alpha(X) = (f_F^\alpha(X'))'$ and substituting we obtain

$$\partial_F^{\alpha+1}(X) = (f_F(f_F^\alpha(X')))' = (f_F^{\alpha+1}(X'))'$$

which is what we wanted.

- Let λ be a limit ordinal and assume that the statement holds for all $\alpha < \lambda$. By definition,

$$\partial_F^\lambda(X) = \bigcap_{\alpha < \lambda} \partial_F^\alpha(X).$$

By De Morgan laws, we have that

$$(\partial_F^\lambda(X))' = \left(\bigcap_{\alpha < \lambda} \partial_F^\alpha(X) \right)' = \bigcup_{\alpha < \lambda} (\partial_F^\alpha(X))'.$$

Using the induction hypothesis, for all $\alpha < \lambda$,

$$(\partial_F^\alpha(X))' = f_F^\alpha(X'),$$

Then,

$$(\partial_F^\lambda(X))' = \bigcup_{\alpha < \lambda} f_F^\alpha(X') = f_F^\lambda(X').$$

Therefore,

$$(\partial_F^\lambda(X))' = f_F^\lambda(X'),$$

that is,

$$\partial_F^\lambda(X) = (f_F^\lambda(X'))'.$$

□

As a particular case, we have the following.

Corollary 5.2. *If $X \in \mathcal{CS}$, $U \in \mathcal{OS}$ and $\alpha = \infty$, then*

$$(7) \quad \begin{aligned} \partial_F^\infty(X) = (v_F(X'))' &\iff \partial_F^\infty(U')' = (v_F(U)) \\ &\iff \partial_F^\infty(X)' = (v_F(X')). \end{aligned}$$

Lemma 5.3. *If $A = \mathcal{OS}$ and $E \subseteq S$, then*

$$\mathcal{U}_*[E]\mathcal{U} = [E].$$

Proof. Since A is spacial, then $\mathcal{U}$ is an isomorphism. Thus, if $U \in \mathcal{O} \text{ pt } A$ exists $V \in \text{pt } A$ such that $U = \mathcal{U}(V)$ and as $U \in \text{pt } A$, then $U = V$. Next,

$$\begin{aligned} (\mathcal{U}_*[E]\mathcal{U})(U) &= \mathcal{U}_*[E](\mathcal{U}(U)) = \mathcal{U}_*([E](U)) \\ &= \mathcal{U}_*(E \cup U)^\circ = \bigcup \{W \in \mathcal{OS} \mid W \subseteq (E \cup U)^\circ\} \\ &= (E \cup U)^\circ = [E](U). \end{aligned}$$

Therefore $\mathcal{U}_*[E]\mathcal{U} = [E]$.

□

Proposition 5.4. . For $F \in A^\wedge$, $Q \in \mathcal{QS}$ and $\nabla = \nabla(Q)$, if $k \in [v_\nabla, w_\nabla]$, then $\mathcal{U}_* \circ k \circ \mathcal{U} \in [v_F, w_F]$.

Proof. If $k \in [v_\nabla, w_\nabla]$, then $\mathcal{U}_* \circ k \circ \mathcal{U} \in NA$. Additionally, for $F \in A^\wedge$ if $j \in [v_F, w_F]$, then $F = \nabla(j)$. In this way we must prove that $F = \nabla(\mathcal{U}_* \circ k \circ \mathcal{U})$.

By Hofmann-Mislove Theorem, we have that

$$x \in F \iff Q \subseteq \mathcal{U}(x) \iff \mathcal{U}(x) \in \nabla = \nabla(Q).$$

Furthermore, for all $x \in A$

$$(\mathcal{U}_* \circ k \circ \mathcal{U})(x) = \mathcal{U}_*(k(\mathcal{U}(x))) = \bigwedge (S \setminus k(\mathcal{U}(x))).$$

Since $k \in [v_\nabla, w_\nabla]$, then $\nabla = \nabla(k)$, that is, for $V \in \nabla$ it holds that $k(V) = S$. Therefore,

$$\begin{aligned} x \in F &\iff \mathcal{U}(x) \in \nabla \iff S \setminus k(\mathcal{U}(x)) = \emptyset \\ &\iff \bigwedge (S \setminus k(\mathcal{U}(x))) = 1 \\ &\iff x \in \nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) \end{aligned}$$

□

We have a morphism $\mathcal{U}: [v_\nabla, w_\nabla] \rightarrow [v_F, w_F]$. Note that this occurs at both ends of the interval on NA .

Take $k \in [v_\nabla, w_\nabla]$. Since v_F is a fitted nuclei, then it holds

$$(8) \quad v_F \leq \mathcal{U}_* \circ k \circ \mathcal{U}.$$

Similarly, as w_F is the largest element in its block,

$$(9) \quad \mathcal{U}_* \circ k \circ \mathcal{U} \leq w_F.$$

Proposition 5.5. If $F \in A^\wedge$ and $\nabla \in \mathcal{OS}^\wedge$, then

$$w_F = \mathcal{U}_* \circ w_\nabla \circ \mathcal{U}.$$

Proof. By (9), we must verify that $w_F \leq \mathcal{U}_* \circ w_\nabla \circ \mathcal{U}$. Equivalently, we must prove that

$$\mathcal{U} \circ w_F \leq w_\nabla \circ \mathcal{U},$$

that is, for all $x \in A$

$$\mathcal{U}(w_F(x)) \subseteq [M'](\mathcal{U}(x)) = (M' \cup \mathcal{U}(x))^\circ.$$

Since $\mathcal{U}(w_F(x)) \in \mathcal{OS}$, it is enough that $\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x)$. As a contradiction, let $y \in \mathcal{U}(w_F(x))$ be such that $y \notin M'$ and $y \notin \mathcal{U}(x)$. Then

$$y \in M \quad \text{and} \quad x \leq y.$$

By hypothesis, $y \in \mathcal{U}(w_F(x))$, then

$$w_F(x) \not\leq y \quad \text{and} \quad w_F(x) = \bigwedge \{m \in M \mid x \leq m\},$$

that is, $w_F(x) \leq y$ and it is a contradiction.

Therefore, $\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x)$. □

Definition 5.6. Let A be a frame.

- (1) A is *strongly stacked* if $\mathcal{U}_*[Q']\mathcal{U} = v_F$.
- (2) A is *stacked* if $(\mathcal{U}_*([Q'])\mathcal{U})(0) = v_F(0)$.

for any (F, Q) HM pair.

Note that strongly stacked implies stacked. Our aim is to prove that the above properties are conservative, that is, S is a stacked (strongly stacked) space if and only if its frame of open sets $\mathcal{O}S$ is stacked (strongly stacked).

Proposition 5.7. *The notions of strongly stacked and stacked are conservative.*

Proof. Consider $S \in \text{Top}$, $A = \mathcal{O}S$, $F \in A^\wedge$ and $Q \in \mathcal{Q}S$.

First, suppose that S is strongly stacked, then $v_F = [Q']$ and, by Lemma 5.3, $\mathcal{U}_*[Q']\mathcal{U} = [Q'] = v_F$. Therefore $\mathcal{O}S$ is strongly stacked.

Now, suppose that S is stacked, then

$$\overline{Q} = \partial_F^\infty(S) \iff \overline{Q}' = (\partial_F^\infty(S))'.$$

Furthermore,

$$\begin{aligned} (\mathcal{U}_*[Q']\mathcal{U})(\emptyset) &= \mathcal{U}_*([Q'](\{V \in \text{pt } \mathcal{O}S \mid \emptyset \not\subseteq V\})) \\ &= \mathcal{U}_*([Q'](\emptyset)) = \mathcal{U}_*(Q')^\circ \\ &= \bigcup \{W \in \mathcal{O}S \mid W \subseteq (Q')^\circ\} \\ &= (Q')^\circ = \overline{Q}'. \end{aligned}$$

By Corollary 5.2, $v_F(\emptyset) = (\partial_F^\infty(\emptyset))'$. Therefore,

$$v_F(\emptyset) = (\mathcal{U}_*[Q']\mathcal{U})(\emptyset),$$

and $\mathcal{O}S$ is stacked. □

6. SEPARATION PROPERTIES IN FRAMES

If $S \in \text{Top}$ is an T_1 and strongly stacked, then it holds that $w_\nabla = v_\nabla$. Additionally, if S is stacked and T_1 , then it is sober.

Thus, we could ask ourselves if the same occurs in frames. In this section we will study the above question.

Take $p \in \text{pt } A$ and $(F_p, \uparrow p)$ a HM pair, then for

$$M = \{m \in A \setminus F_p \mid (\forall a \in A \setminus F_p)[a \leq m]\},$$

the nuclei $w_{F_p}(x) = \bigwedge \{m \in M \mid x \leq m\}$ is $w_p(x)$.

To prove this, note that $A \setminus F_p = \{x \in A \mid x \leq p\} = \downarrow p$. As,

$$M = \{m \in \downarrow p \mid [\forall a \in \downarrow p](a \leq m)\} = \{p\}.$$

Hence, $w_{F_p}(x) = \bigwedge \{p \mid x \leq p\}$, that is,

$$(10) \quad w_{F_p}(x) = \begin{cases} 1, & \text{if } x \not\leq p \\ p, & \text{if } x \leq p \end{cases} = w_p(x)$$

for all $x \in A$. Therefore $w_{F_p} = w_p$.

Proposition 6.1. *Let $A \in \mathbf{Frm}$ and (F, Q) be any HM pair. If $v_F(0) = w_F(0)$, then for all $p \in \text{pt } A$, p is maximal in $\text{pt } A$.*

Proof. Consider $p, q \in \text{pt } A$ such that $p \leq q$. Note that for $q \in \text{pt } A$, we have $(F_q, \uparrow q)$ a HM pair, with

$$F_q = \{x \in A \mid \uparrow q \subseteq \mathcal{U}(x)\} = \{x \in A \mid (\forall r \in \text{pt } A)[r \leq q \Rightarrow x \not\leq r]\}.$$

By hypothesis, $v_{F_q}(0) = w_{F_q}(0)$ and by (10)

$$q = w_q(0) = w_{F_q}(0) = v_{F_q}(0) \leq v_{F_q}(p) = p.$$

Therefore, $p = q$ and every point is maximal in $\text{pt } A$. $\square$

As we show later on (2), for all (F, Q, ∇) HM tupla we have

$$\mathcal{U} \circ v_F \leq v_\nabla \circ \mathcal{U}.$$

Additionally, if A is strongly stacked, then $v_F = \mathcal{U}_*[Q']\mathcal{U}$ and

$$\mathcal{U}_* \circ v_\nabla \circ \mathcal{U} \leq \mathcal{U}_* \circ [Q']\mathcal{U}$$

Thus, $v_F = \mathcal{U}_* \circ v_\nabla \circ \mathcal{U}$, equivalently $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$.

Therefore, if A is strongly stacked, then the diagram (2) is commutative. The following result tells us under which conditions we have the converse implication.

Proposition 6.2. *If $A \in \mathbf{Frm}$ is tidy and the diagram*

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A \\ \mathcal{U} \downarrow & & \downarrow \mathcal{U} \\ \mathcal{OS} & \xrightarrow{v_\nabla} & \mathcal{OS} \end{array}$$

commutes, then A is strongly stacked.

Proof. Let $A \in \mathbf{Frm}$, $S = \text{pt } A$ and (F, Q) be any HM pair. By hypothesis, A is tidy and by previous results we have the implications

$$A \text{ tidy} \Rightarrow \mathcal{OS} \text{ tidy} \Rightarrow S \text{ stacked and packed}.$$

Since S is packed, if $Q \in \mathcal{QS}$, then $Q \in \mathcal{CS}$ and $Q' \in \mathcal{OS}$. Then, as S is stacked, we have that $\overline{Q} = \partial_\nabla^\infty(S)$, that is, $Q' = v_\nabla(\emptyset)$ and as $\mathcal{OS}$ is tidy, $v_\nabla = u_{Q'} = [Q']$.

Additionally, if $v_\nabla \circ \mathcal{U} = \mathcal{U} \circ v_F$, then

$$v_\nabla \circ \mathcal{U} \leq \mathcal{U} \circ v_F \quad \text{and} \quad \mathcal{U} \circ v_F \leq v_\nabla \circ \mathcal{U},$$

that is,

$$[Q'] \circ \mathcal{U} \leq \mathcal{U} \circ v_F \quad \text{and} \quad \mathcal{U} \circ v_F \leq [Q'] \circ \mathcal{U}.$$

By properties of the right adjoint

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq v_F \quad \text{and} \quad v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore $v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$ and A is strongly stacked. $\square$

Consequently, we have the following result.

Corollary 6.3. *If A is tidy, then $S = \text{pt } A$ is strongly stacked.*

Proof. Since S is strongly stacked if and only if $v_\nabla = [Q']$, the result follows from the fact that

$$\mathcal{O}S \text{ tidy} \Rightarrow S \text{ strongly stacked}$$

$\square$

With the morphism $\mathcal{U}_A$, we have the nuclei sp_A defined by

$$\text{sp}_A(x) = \bigwedge \{p \in \text{pt } A \mid x \leq p\}$$

for $x \in A$. Combining this fact with the Proposition 6.1 we know information about the collapse on admisibility interval.

Proposition 6.4. *For $A \in \text{Frm}$ and (F, Q) any HM pair, the following statements hold:*

- (1) *If A is T_1 and stacked, then $v_F(0) = w_F(0)$.*
- (2) *If $v_F(0) = w_F(0)$, then A is stacked.*
- (3) *If $v_F(0) = w_F(0)$ and the set $Y = \{q \in \text{pt } A \mid a \leq q\}$ is non-empty for all $a \in A$, then A is T_1 .*

Proof.

- (1) If A is T_1 , then

$$\mathcal{O}S \text{ is } T_1 \Rightarrow S \text{ is } T_1 \Rightarrow Q = M.$$

Since A is stacked,

$$v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) = (\mathcal{U}_* \circ [M'] \circ \mathcal{U})(0) = w_F(0).$$

- (2) Recall that for every HM pair, the following inequality holds:

$$(11) \quad v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F$$

and by hypothesis $v_F(0) = w_F(0)$, that is, $v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) = w_F(0)$. Therefore, A is stacked.

- (3) Consider $p \in \text{pt } A$ and $a \in A$ such that $p \leq a$. By hypothesis $\{q \in \text{pt } A \mid a \leq q\} \neq \emptyset$. Then

$$p \leq a \leq \text{sp}(a) = \bigwedge \{q \in \text{pt } A \mid a \leq q\}$$

for some $q \in Y$. Thus, $p \leq a \leq q$ and by Proposición 6.1, $p = q$. Therefore, $p = a$ and every point is maximal in A , that is, A is T_1 . $\square$

Corollary 6.5. *For $A \in \mathbf{Frm}$ and (F, Q) any HM pair, the following statements hold:*

- (1) *If A is T_1 and strongly stacked, then $v_F = w_F$.*
- (2) *If $v_F = w_F$, then A is strongly stacked.*
- (3) *If $v_F = w_F$ and the set $Y = \{q \in \text{pt } A \mid a \leq q\}$ is non-empty for all $a \in A$, then A is T_1 .*

Proof.

- (1) If A is strongly stacked, then $v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$ and by T_1 we have that $Q = M$. Therefore $v_F = w_F$.
- (2) This follows from (11).
- (3) This follows from Proposition 6.4.

□

Note that, by 6.1 and 6.4, if $v_F(0) = w_F(0)$, then A is stacked and every point is maximal in $\text{pt } A$. Particularly, for each $p \in \text{pt } A$ and $(F_p, \uparrow p)$ it holds $v_{F_p}(0) = p$. Furthermore,

$$v_{F_p}(p) = v_{F_p}(0) = 0_{A_{F_p}}.$$

If A is T_1 , p is maximal in A , then p is maximal in A_{F_p} and $A_{F_p} = \{q, 1\}$.

Now, $F_{F_p} = \nabla(v_{F_p})$ and $x \in F_p$ if and only if $v_{F_p}(x) = 1$. In this way, under the previous hypotheses, it holds that

$$v_{F_p}(x) = \begin{cases} 1, & \text{si } x \not\leq p \\ p, & \text{si } x \leq p \end{cases} = \begin{cases} 1, & \text{si } x \in F_p \\ p, & \text{si } x \notin F_p \end{cases} = w_{F_p}(x).$$

Therefore, if A is T_1 and stacked, then $v_{F_p} = w_{F_p}$ for each $p \in \text{pt } A$. With this in mind, we could ask ourselves if there exists some way to recover the notion of strongly stacked by means of T_1 and stacked.

Now, we need the following information:

- (1) Let $A^\wedge$ be a topological space where the basic open sets of $\mathcal{O}(A^\wedge)$ are given by

$$D(x) = \{F \in A^\wedge \mid x \in F\}$$

where $x \in A$.

- (2) For $A \in \mathbf{Frm}$, we can construct a frame VA as in [Sim04]. Each element in $\text{pt}(VA)$ is of the form

$$(F, a), \quad \text{where } F \in A^\wedge \text{ y } a \in [v_F(0), w_F(0)].$$

- (3) Take a continuous surjective map $\pi: \text{pt}(VA) \rightarrow A^\wedge$, defined by

$$(F, a) \mapsto F.$$

- (4) By the functor $\mathcal{O}(-)$ we have

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \rightarrow \mathcal{O}(\text{pt}(VA)).$$

(5) Open sets in $\text{pt}(VA)$ are given by the following:

$$(12) \quad (F, a) \in [-](x) \iff x \in F \quad \text{and} \quad (F, a) \in \langle - \rangle(x) \iff x \not\leq a.$$

Particularly,

$$(13) \quad a \in [F](x) \iff x \in F \quad \text{and} \quad a \in \langle F \rangle(x) \iff x \not\leq a.$$

Proposition 6.6. *Let $A, VA \in \text{Frm}$ and $A^\wedge$ be as above. The following statements hold:*

- (1) *If A is T_1 and stacked, then $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$ under the continuous function π .*
- (2) *If the morphism $\mathcal{O}(\pi) = \pi^{-1}$ induces that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$, then A is stacked.*
- (3) *If the morphism $\mathcal{O}(\pi) = \pi^{-1}$ induces that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$ and*

$$\mathcal{V}(a) = \{q \in \text{pt } A \mid a \leq q\} \neq \emptyset,$$

then A is T_1 .

Proof.

- (1) If A is T_1 and stacked, then for each $F \in A^\wedge$ it holds that

$$v_F(0) = w_F(0).$$

Take $(F, a), (G, b) \in \text{pt}(VA)$ such that $\pi(F, a) = F = \pi(G, a) = G$. By T_1 and stacked we have

$$a = v_F(0) = v_G(0) = b.$$

Therefore $(F, a) = (G, b)$, that is, π is injective. Since it is also surjective, π is bijective.

We must show that $\pi^{-1}: A^\wedge \rightarrow \text{pt}(VA)$ is continuous. For this, recall that by (12)

$$\begin{aligned} (F, a) \in [-](x) &\iff x \in F \\ &\iff (F, a) \in \pi^{-1}(D(x)) \\ &\iff \pi([-](x)) = D(x). \end{aligned}$$

For the other basic open sets, by (13) we have

$$a \notin \langle F \rangle(x) \iff x \leq a \iff v_F(x) \leq a.$$

By T_1 and stacked we have that $a = v_F(0)$ and thus

$$v_F(0) \notin \langle F \rangle(x) \iff x \leq v_F(0) \iff v_F(x) = v_F(0).$$

As $v_F(0) \neq 1$,

$$v_F(0) \notin \langle F \rangle(x) \iff x \notin F$$

or equivalently

$$a \in \langle F \rangle(x) \iff (F, a) \in \langle - \rangle(x) \iff x \in F.$$

Therefore, $\pi(\langle - \rangle(x)) = D(x)$ y π^{-1} is continuous. Hence, π is a homeomorphism and

$$\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA)).$$

- (2) Suppose that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$ under $\mathcal{O}(\pi) = \pi^{-1}$. Since π^{-1} is bijective, then π is also bijective. Thus, if $F \in A^\wedge$, then

$$(F, v_F(0)), (F, w_F(0)) \in \text{pt}(VA)$$

and

$$\pi(F, v_F(0)) = F = \pi(F, w_F(0)).$$

By the injectivity of π we have that $v_F(0) = w_F(0)$ and therefore A is stacked.

- (3) Suppose that $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$ and A is stacked. Consider $a \in A$ and $p \in \text{pt } A$ such that $p \leq a$. Moreover, by the previous item we have that

$$\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA)) \Rightarrow v_F(0) = w_F(0)$$

and every point is maximal in $\text{pt } A$. Since $\mathcal{V}(a) \neq \emptyset$, there exists $q \in \text{pt } A$ such that $a \leq q$. By the maximality of p and q , we have

$$p \leq a \leq q = p.$$

Therefore, $p = a$ and A is T_1 .

□

Note that item (3) holds whenever A is spatial. In the spatial setting, this corresponds to the topological fact that

$$S \text{ is } T_1 \text{ and stacked} \Rightarrow S \text{ is sober.}$$

However, the point-free setting allows for a potentially broader phenomenon, since the conditions of being T_1 and stacked do not a priori force a frame to be spatial. This leads naturally to the following question:

Do there exist T_1 stacked frames that are not spatial?

7. QUOTIENTS OF STACKED FRAMES

Remember that for $(f: A \rightarrow B) \in \text{Frm}$, $j \in NA$ and $k \in NB$, we have the following definitions:

$$N(f)(j) = \bigvee \{u_{f(j(a))} \wedge v_{f(a)} \mid a \in A\} \in NA_j$$

and

$$N(f)_*(k) = f_* \circ k \circ f \in NA,$$

where f_* is the right adjoint of f .

Proposition 7.1. *Let A be a frame and $j \in NA$. If A is strongly stacked, then A_j is strongly stacked.*

Proof. Consider $S = \text{pt } A$, $S_j = \text{pt } A_j$ and let $F \in A_j^\wedge$ be. We define:

$$G = j_*(F), \quad Q = S_j \setminus F, \quad R = S \setminus G,$$

where j_* is the right adjoint associated with the frame morphism $j^*: A \rightarrow A_j$ and $x \in G$ if and only if $j(x) \in F$. Considering the spatial reflections

$$\mathcal{U}: A \rightarrow \mathcal{O}S \quad \text{and} \quad \mathcal{U}_j: A_j \rightarrow \mathcal{O}S_j$$

we have the following diagram

$$\begin{array}{ccccc}
 & & A & \xrightarrow{\mathcal{U}} & \mathcal{O}S \\
 & \nearrow v_G & \downarrow j^* & \nearrow [R'] & \downarrow sj \\
 A & \xrightarrow{\mathcal{U}} & \mathcal{O}S & & A_j \xrightarrow{\mathcal{U}_j} \mathcal{O}S_j \\
 \downarrow j^* & & \downarrow sj & \nearrow v_F & \downarrow [Q'] \\
 A_j & \xrightarrow{\mathcal{U}_j} & \mathcal{O}S_j & &
 \end{array}$$

Using the right adjoints of $\mathcal{U}$ and $\mathcal{U}_j$ we obtain

$$v_F, \quad l = N(\mathcal{U}_j)_*([Q']), \quad m = N(j^*)(N(\mathcal{U})_*([R'])), \quad n = N(j^*)(N(\mathcal{U}_j \circ j^*)_*)([Q'])$$

are nuclei in NA_j . Let us see what is the relationship between these nuclei.

By construction, we have $v_F \leq N(\mathcal{U}_j)_*([Q'])$ and we must verify the other inequality for A_j to be strongly stacked.

By hypothesis, we have $v_G = N(\mathcal{U})_*([R'])$. Moreover, for $x \in A$, we have $j(v_x) \leq v_{j(x)}$ and, by definition of the adjusted nuclei associated with F and G ,

$$(14) \quad j \circ v_G \leq v_F.$$

Furthermore, for $(j: A \rightarrow A_j) \in \text{Frm}$

$$N(j^*)(v_G) = \bigvee \{u_{j(v_G(a))} \wedge v_{j(a)} \mid a \in A\} \quad \text{y} \quad v_F = \bigvee \{u_{v_F(a)} \wedge v_a \mid a \in A_j\}.$$

Note, by (14), $u_{j(v_G)} \leq u_{v_F}$ and therefore

$$(15) \quad m \leq v_F.$$

To l and n , we have

$$n = \bigvee \{u_{j^*(n(x))} \wedge v_{j^*(x)} \mid x \in A\} \quad \text{and} \quad l = \bigvee \{u_{l(x)} \wedge v_x \mid x \in A_j\},$$

where

$$j^*(n(x)) = j^*((j_* \circ \mathcal{U}_{j*} \circ [Q'] \circ \mathcal{U}_j \circ j^*)(x)) \quad \text{y} \quad l(x) = N(\mathcal{U}_j)_*([Q'])(x).$$

Then,

$$\begin{aligned} j^*(n(x)) &= j^*(j_*(\bigvee \{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\})) \\ &= \bigvee \{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\}. \end{aligned}$$

Since $x \in A_j$,

$$l(x) = \bigvee \{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\}.$$

Therefore $j(n(x)) = l(x)$.

To $m(x)$ we have

$$m(x) = \bigvee \{u_{j^*}(\mathcal{U}_* \circ [R'] \circ \mathcal{U})(x) \wedge v_{j^*}(x) \mid x \in A\}.$$

Since $A_j \subseteq A$, then $Q = R \cap A_j$ and

$$m(x) = \bigvee \{u_{j^*}(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(x) \wedge v_x \mid x \in A_j\}.$$

Note

$$\begin{aligned} j^*(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(x) &= j^*(\bigvee \{a \in A \mid \mathcal{U}(a) \subseteq Q' \cup \mathcal{U}(x)\}) \\ &= \bigvee \{j^*(a) \in A_j \mid \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\}. \end{aligned}$$

To $x \in A_j$

$$\begin{aligned} \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(j^*(x)) &\iff \mathcal{U}(j^*(a)) \cap S_j \subseteq (Q' \cup \mathcal{U}(j^*(x))) \cap S_j \\ &\iff \mathcal{U}_j(j^*(a)) \subseteq Q' \cup \mathcal{U}_j(j^*(x)), \end{aligned}$$

that is,

$$\bigvee \{j^*(a) \in A_j \mid \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\} = j^*(N(\mathcal{U}_j \circ j)_*)([Q']).$$

Thus, we have

$$\bigvee \{u_{j^*}(N(\mathcal{U}_j \circ j)_*)([Q'])(x) \wedge x \mid x \in A\} = m(x).$$

Since $(N(\mathcal{U}_j \circ j)_*)([Q']) \leq j(N(\mathcal{U}_j \circ j)_*)([Q'])$ and by (15)

$$l \leq m \leq v_F \leq l.$$

Therefore $v_F = \mathcal{U}_{j^*} \circ [Q'] \circ \mathcal{U}_j$, that is, A_j is strongly stacked. $\square$

Since every strongly stacked frame is stacked, we obtain the following result.

Corollary 7.2. *Let A be a frame and $j \in NA$. If A is strongly stacked, then A_j is stacked.*

Proof. By the previous proposition, if A is strongly stacked, then A_j is strongly stacked and this implies that A_j is stacked. $\square$

8. RELATIONSHIP TO THE PROPERTY (H)

If A is a Hausdorff frame, then A is T_1 . In this way, to be able to use the example of Paseka and Smarda it is enough that (H) implies stacked.

Proposition 8.1. *If $A \in \text{Frm}$ satisfies (H), then A is stacked.*

Proof. If A satisfies (H), then A is T_1 , that is, for every pair of points $x, y \in A$ with $x \neq y$, there exist neighborhoods U, V of x, y respectively such that $U \cap V = \emptyset$. This implies that

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F,$$

that is, $\mathcal{U}_* \circ [Q'] \circ \mathcal{U}(0) = w_F(0)$ and since $w_F \neq \text{tp}$, we have $w_F(0) \neq 1$.

By construction, we have that $v_F(0) \leq w_F(0)$ and thus, we need to verify that the reverse inequality holds to conclude that A is stacked. For a contradiction, suppose that

$$w_F(0) \not\leq v_F(0).$$

Since $1 \neq w_F(0) \not\leq v_F(0)$, by (H), exists $c \in A$ such that

$$c \not\leq w_F(0) \quad \text{y} \quad \neg c \not\leq v_F(0).$$

If $\neg c \not\leq v_F(0)$, then $c \notin F$ □

9. MARCOS KC

Definition 9.1. Let A be a frame. We say that A is:

- (1) *KC* if each compact quotient is closed.
- (2) *Hausdorff closed compact* (or *KCH* for short) if each compact quotient of A is closed and Hausdorff.

La Definición 9.1 nos dice que si $j \in NA$, entonces $\nabla(j) \in A^\wedge$ y $j = u_a$ para algún $a \in A$. De manera adicional, para 2) pedimos que A_j cumpla la propiedad (H).

Proposition 9.2. *If $A \in \text{Frm}$ satisfies KC, then A_j satisfies KC for each $j \in N(A)$.*

Proof. Consider $k \in NA_j$ such that $(A_j)_k$ is compact. If $(A_j)_k$ is compact, then $\nabla(k) \in A_j^\wedge$ and, by Proposition 3.1, $j_* \nabla(k) \in A^\wedge$.

Let $l = j_* \circ k \circ j^* \in NA$ be, then A_l is a compact quotient of A and there exists $a \in A$ such that $l = u_a$. Thus

$$\begin{array}{ccccccc} & & & l & & & \\ & & & \curvearrowright & & & \\ A & \xrightarrow{j^*} & A_j & \xrightarrow{k} & (A_j)_k & \xrightarrow{j_*} & A_j \subseteq A \end{array}$$

and $a \vee x = k(j(x))$. Therefore, if $x = a$, $k(j(x)) = a$.

We need that $k = u_b$ for some $b \in A_j$. If $x \in A_j$ and $b = j(a)$

$$\begin{aligned} u_b(x) &= b \vee x = b \vee j(x) = j(j(a) \vee j(x)) \\ &= j(k(j(a)) \vee x) \\ &= j(u_a(x)) \\ &= j(k(x)) \\ &= k(x). \end{aligned}$$

Hence $u_b = k$. □

Proposition 9.3. *If A is a KC frame, then A is a T_1 frame.*

Proof. Let $p \in \text{pt } A$ and $a \in A$ such that $p \leq a \leq 1$. Consider

$$w_p(x) = \begin{cases} 1 & \text{si } x \not\leq p \\ p & \text{si } x \leq p \end{cases}$$

to $x \in A$. $P = \nabla(w_p) = \{x \in A \mid x \not\leq p\}$ is a completely prime filter (in particular, $P \in A^\wedge$). Because A is KC , then A_{w_p} is a compact closed quotient. Thus $u_p = w_p$ and

$$u_p(a) = a \quad \text{y} \quad w_p(a) = 1.$$

that is, $a = 1$. Therefore p is maximal. □

One of our aims is study nuclei such that produce compact quotient, particularly, nuclei that produce compact closed quotient.

Define the following subset of A

$$uA = \{a \in A \mid A_{u_a} \text{ es un cociente compacto}\}.$$

Note that, to $a \in uA$ we have $\nabla(u_a) \in A^\wedge$, that is, define the map

$$\alpha: uA \rightarrow A^\wedge.$$

Let us see the properties of this map.

Lemma 9.4. *For $uA \subseteq A$ as before, the following statements hold:*

- (1) *If A is tidy, then α is surjective.*
- (2) *If u_a is fitted for all $a \in uA$, then α is injective.*
- (3) *If A is tidy and $a \in uA$, then α is bijective if and only if u_a is fitted.*

Proof.

- (1) Consider $F = \nabla(j) \in A^\wedge$ and v_F the fitted nucleus associated with F . If A is tidy, then $v_F = u_d$, where $d = v_F(0) \in A$. Hence,

$$\alpha(d) = \nabla(u_d) = \nabla(v_F) = F.$$

and α is surjective.

- (2) Consider $a, b \in uA$ such that $\alpha(a) = \alpha(b)$, that is, $\nabla(u_a) = \nabla(u_b)$. By hypothesis, u_a, u_b are fitted nuclei, that is,

$$u_c \leq u_d \quad \text{y} \quad u_c \geq u_d.$$

Evaluating both nuclei at 0 we get $c = d$ and therefore α is injective.

- (3) Let A be tidy, $a \in uA$ and suppose that α is bijective. Consider $F = \nabla(u_a)$ and v_F the fitted nucleus associated with F . By efficiency, $v_F = u_d$, that is,

$$\alpha(a) = \nabla(u_a) = \nabla(v_F) = \nabla(u_d) = \alpha(d)$$

where $d = v_F(0)$.

By hypothesis, α is bijective, then, by injectivity, $a = d$. Therefore $v_F = u_a$ and u_a is fitted.

Conversely, since A is tidy, then by 1), α is surjective. By hypothesis, u_a is fitted and, by 2), α is injective. Therefore α is bijective. □

Using KC frames, we have the following result.

Corollary 9.5. *Let A be a KC frame and $F \in A^\wedge$. Then α is bijective if and only if for all $j \in [v_F, w_F]$, j is fitted.*

Proof. Suppose that A is a KC frame and let $F \in A^\wedge$. By definition, for all $j \in [v_F, w_F]$, we have $j = u_a$ for some $a \in A$. Since A is a KC frame, it is efficient and by the previous lemma, α is bijective if and only if for all $j \in [v_F, w_F]$, j is fitted. □

Proposition 9.6. *Let A be a frame. A is KC if and only if A is tidy and $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$.*

Proof.

$\Rightarrow$): Suppose that A is a KC frame. Trivially, KC implies efficient and thus, it remains to verify that KC implies that $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$ or equivalently, by properties of the right adjoint,

$$v_F = \mathcal{U}_* \circ v_\nabla \circ \mathcal{U}.$$

If A is a KC frame, then for any $F \in A^\wedge$, it holds that for all $j \in [v_F, w_F]$, $j = u_a$ for some $a \in A$. In particular,

$$v_F = u_d, \quad \text{y} \quad \mathcal{U}_* \circ v_\nabla \circ \mathcal{U} = u_a$$

where $d, a \in A$.

By construction, always holds that $u_d \leq u_a$. To see the other inequality, we verify that the function α is injective. Note that $d \leq a$

$\Leftarrow$): Suppose that A is efficient and $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$ for all HM triples (F, Q, ∇) . First, if A is efficient, then A is T_1 . Moreover, if A is efficient and $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$, then A is strongly stacked. Therefore, A is T_1

and strongly stacked. The above implies that $v_F = w_F$ and, by efficiency, $v_F = u_d$, where $d = v_F(0)$. Thus, for all $j \in [v_F, w_F]$, $j = u_d$, meaning that A is KC .

□

10. EXAMPLES

10.1. The frame $[0, 1]$. Following the idea presented in [Sim06b] (or, alternatively, in [Sim04]), let us consider the linearly ordered frame

$$A = [0, 1].$$

In this case the following observations hold.

- (1) For all $a, b \in A$ the implication $(a \succ b)$ is given by

$$(a \succ b) = \begin{cases} 1, & \text{si } a \leq b, \\ b, & \text{si } a > b. \end{cases}$$

- (2) For $a, x \in A$, the distinguished nuclei u_a, v_a, w_a are given by

$$u_a(x) = \begin{cases} x & \text{si } a \leq x, \\ a & \text{si } x < a, \end{cases}, \quad v_a(x) = \begin{cases} 1 & \text{si } x \geq a, \\ a & \text{si } x < a, \end{cases}, \quad w_a(x) = \begin{cases} 1 & \text{si } x > a, \\ a & \text{si } x \leq a. \end{cases}$$

- (3) If $F \subseteq A$ is a filter, then F is always admissible and has the form

$$F = \nabla(w_a) = (a, 1] \quad \text{o} \quad F = \nabla(v_a) = [a, 1]$$

for some $a \in A$. In particular, $F \in A^\wedge$ if and only if $F = (a, 1]$.

- (4) If $F \in A^\wedge$, the adjusted nucleus associated v_F is calculated as

$$(16) \quad v_F(x) = l_m(x) = \begin{cases} 1 & \text{if } x > m, \\ x & \text{if } x \leq m. \end{cases}$$

where $l_m = v_m \wedge w_m$ and $m = \bigwedge F$.

- (5) For $S = \text{pt } A$ it holds that $S = [0, 1)$ and the open sets in $\mathcal{O}S$ are of the form

$$\mathcal{U}_A(x) = \{p \in S \mid p < x\} = [0, x).$$

- (6) The frame A is not T_1 , since no $p \in S$ is maximal. Consequently, A does not satisfy any stronger separation property in Frm .

- (7) For $F \in A^\wedge$ the compact saturated set associated $Q \in \mathcal{Q}S$ is of the form $Q = [0, m]$.

- (8) It holds that $[Q'] = v_F$; therefore, S is strongly stacked.

- (9) Since $[0, x) \in \mathcal{O}S$, then $[x, 1] \in \mathcal{C}S$ and $Q = [0, m]$. Therefore, the compact subsets are not closed; that is, S is not packed.

- (10) The frame A is not efficient.

The above resume some characteristics about the frame $[0, 1]$. Now, we will perform the analysis of the patch constructions of A and its point space S .

Remember that for $A \in \text{Frm}$ and $S \in \text{Top}$ we have

$$\text{Pbase} = \{u_a \wedge v_F \mid a \in A, F \in A^\wedge\} \quad \text{y} \quad \text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}$$

where Pbase generates the patch frame of A and pbase generates the topology of the patch space of S . By (16) we have that

$$\text{Pbase} = \{u_a \wedge l_m \mid a \in [0, 1], m \in [0, 1]\}.$$

For $x \in A$ we have

$$(u_a \wedge l_m)(x) = u_a(x) \wedge l_m(x).$$

Let us analyze the value of this map according to the relative position of x and m .

Case 1: If $x < m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge x = x.$$

Case 2: If $x = m$, then

$$(u_a \wedge l_m)(x) = u_a(m) \wedge m = x.$$

Case 3: If $x > m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge 1 = u_a(x).$$

In the first two cases, the value of a does not affect the evaluation. Thus, the only non-trivial case occurs when $m < x$, where the value of $(u_a \wedge l_m)(x)$ depends on the relative position of a and x . We therefore distinguish the following two subcases.

Case 3.1: If $m < x < a$, then

$$(u_a \wedge l_m)(x) = a.$$

Case 3.2: If $m < a \leq x$, then

$$(u_a \wedge l_m)(x) = x.$$

From the preceding analysis, we conclude that the non-trivial generators of Pbase correspond precisely to the pairs (m, a) with $m < a$. Consequently, the frame PA is determined by the open intervals of the form (m, a) , where $m \in \text{pt } A$ and $a \in A$.

We now turn to the question of whether PA satisfies any separation property. To this end, we recall the relationship between the point-free patch construction and its spatial counterpart.

Let A be an arbitrary frame and let $S = \text{pt } A$ be its space of points. The two patch constructions are related by the following commutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & PA & \hookrightarrow & NA \\ \mathcal{U}_A \downarrow & & \downarrow P(\mathcal{U}_A) & & \downarrow N(\mathcal{U}_A) \\ \mathcal{O}S & \longrightarrow & P\mathcal{O}S & \hookrightarrow & N\mathcal{O}S \\ & \searrow & \downarrow \pi & & \downarrow \sigma \\ & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S \end{array}$$

where $\mathcal{O}^f S$ is the Skula topology (or front topology) of S . In [SS06b] is named *the patch assembly diagram* and more details about it can also be found in [Sex03].

In our case, $\mathcal{U}_A$ is an isomorphism. Moreover, under $\mathcal{U}_A$, the patch construction turns out to be functorial and, therefore, $PA \cong POS$. Similarly, $NA \cong NOS$. To describe explicitly POS and NOS , we use the following result, which can be consulted in [ÁBMZ20].

Proposition 10.1. *Let S be a partially ordered set.*

- (1) *If S has no infinite antichains, then NOS is spatial.*
- (2) *If S is totally ordered, then NOS is spatial.*

Thus, by Proposition 10.1 we have that NOS is spatial and therefore $NOS \cong \mathcal{O}^f S$. With all this information, the patch assembly diagram, in our particular case is as follows.

$$\begin{array}{ccccc} A & \longrightarrow & PA & \longrightarrow & NA \\ \cong \downarrow & & \downarrow \cong & & \downarrow \cong \\ \mathcal{O}S & \longrightarrow & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

Therefore, if we want to study the frame PA , we can study the patch topology $\mathcal{O}^p S$. Remember that

$$\text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}.$$

In our case,

$$\text{pbase} = \{[0, a) \cap [m, 1) \mid a \in [0, 1], m \in [0, 1)\} = \{(m, a)\}.$$

Since the topology generated by the intervals (m, a) coincides with the usual topology on $[0, 1)$, the space ${}^p S$ is Hausdorff. Consequently, $\mathcal{O}^p S \cong PA$ satisfies **(H)**, even when A is not T_1 nor tidy.

Lemma 10.2. *Let A be a frame, if the morphism $pU: pA \rightarrow p\mathcal{O}S$ is an isomorphism, then A is strongly stacked.*

Proof. Consider any open filter F on A and the compact saturated determined by it, say Q , thus we have the nuclei v_F and $U_*[Q']U = Q[U]$, we need to observe the action of pU on these two nuclei, in order to do that remember that we have a commutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & pA & \xhookrightarrow{\iota} & NA \\ \mathcal{U}_A \downarrow & & \downarrow p(\mathcal{U}_A) & & \downarrow N(\mathcal{U}_A) \\ \mathcal{O}S & \longrightarrow & p\mathcal{O}S & \xhookrightarrow{\iota} & NOS \\ & \searrow & \downarrow \pi & & \downarrow \sigma \\ & & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

since $N(\mathcal{U})\iota = v\mathcal{U}$, that is, $p\mathcal{U}$ is the restriction of $N(\mathcal{U})$ to pA , then

$$\text{p}\mathcal{U}(v_F) = N(\mathcal{U})(v_F) = \bigvee \{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism $\mathcal{U}$ transforms open filters we have

$$\bigvee \{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee \{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus $v_{\mathcal{U}F}$.

□

REFERENCES

- [ÁBMZ20] Francisco Ávila, Guram Bezhanishvili, PJ Morandi, and Angel Zaldívar, *The frame of nuclei on an alexandorff space*, Order (2020).
- [Hoc69] Melvin Hochster, *Prime ideal structure in commutative rings*, Transactions of the American Mathematical Society **142** (1969), 43–60.
- [Joh82] P. T. Johnstone, *Stone spaces*, Cambridge Studies in Advanced Mathematics, vol. 3, Cambridge University Press, Cambridge, 1982. MR 698074
- [PP12] J. Picado and A. Pultr, *Frames and locales: Topology without points*, Frontiers in Mathematics, Springer Basel, 2012.
- [PP21] Jorge Picado and Aleš Pultr, *Separation in point-free topology*, Springer, 2021.
- [Sex03] Rosemary A Sexton, *A point-free and point-sensitive analysis of the patch assembly*, The University of Manchester (United Kingdom), 2003.
- [Sim04] Harold Simmons, *The vietoris modifications of a frame*, Unpublished manuscript, 79pp., available online at <http://www.cs.man.ac.uk/hsimmons> (2004).
- [Sim06a] ———, *The assembly of a frame*, University of Manchester (2006).
- [Sim06b] ———, *Regularity, fitness, and the block structure of frames*, Applied Categorical Structures **14** (2006), 1–34.
- [SS06a] RA Sexton and H Simmons, *An ordinal indexed hierarchy of separation properties*, to appear (2006).
- [SS06b] ———, *Point-sensitive and point-free patch constructions*, Journal of Pure and Applied Algebra **207** (2006), no. 2, 433–468.